Thin Film Deposition by Thermal Evaporation

Investigating Physical Vapor Deposition

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Abstract

This experiment demonstrates deposition of a Cu film onto a glass substrate using resistive thermal evaporation, a physical vapor deposition (PVD) technique. The charge material, copper, was loaded into a tungsten boat inside a high-vacuum chamber evacuated sequentially by a rotary backing pump and a turbo-molecular pump to a base pressure better than 3×10^{-5} mbar. Resistive heating of the boat, controlled via a variac, raised the local vapor pressure of the charge material until evaporation occurred; the vapor traversed the chamber along a collision-free line of sight (mean free path exceeding the source-to-substrate distance) and condensed on a glass slide substrate. Deposition rate and film thickness were monitored in real time using a quartz crystal microbalance interfaced with STM2 software. The vacuum sequence, current ramp rate, and shutter timing were controlled to obtain a film of the target thickness while minimizing contamination and thermal stress on the source. The measured deposition rate, final film thickness, and associated uncertainties are reported and compared against the expected values, with a brief discussion of the dominant sources of error in the vacuum and thickness-monitoring subsystems.

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1 Introduction

Thin film deposition is the process of applying a layer of material, ranging in thickness from a few nanometers to about 100 µm, onto a substrate surface or onto a previously deposited layer to build up a multilayer structure. The technique finds applications in microfabrication and semiconductor device processing, where controlled, uniform, and reproducible film growth is essential.

Several physical vapor deposition (PVD) methods are used to produce thin films, including sputtering, pulsed laser deposition, and electron beam epitaxy. This experiment employs *thermal evaporation*, in which a solid charge material is resistively heated in vacuum until it evaporates, and the resulting vapor condenses on a substrate to form a film. Thermal evaporation is useful due to its comparative simplicity, low cost, and the reasonable growth rates it offers for metals with moderate melting points.

The aims of this experiment are:

- To evacuate a deposition chamber to high vacuum using a two-stage rotary + turbomolecular pumping system, following the correct valve sequence.
- To deposit a thin metallic film of Cu on a glass substrate by resistive thermal evaporation, and to record the deposition rate and film thickness in situ using a quartz crystal monitor.

2 Theory

2.1 Physical Vapor Deposition by Thermal Evaporation

Thermal evaporation is a PVD technique in which a source material is heated to a temperature at which its equilibrium vapor pressure causes net evaporation from the exposed surface. Mean free path:

$$\lambda = \frac{k_B T}{\sqrt{2}\,\pi d^2 P} \tag{1}$$

where k_B is Boltzmann constant, T is gas temperature, d is effective molecular diameter of the residual gas, and P is chamber pressure. If the chamber is evacuated to sufficiently high vacuum, λ of the evaporated particles exceeds the source-to-substrate distance. Under this condition they travel in straight, collision-free trajectories, and the film grows by condensation with minimal scattering losses or gas-phase reactions. This is the why a base vacuum on the order of 10^{-5} mbar is required before deposition.

2.2 Source-to-Substrate Distance

In our system, the source-to-substrate distance was measured to be $d_{ss} = 15$ cm. At the turbo-pumped base pressure of the chamber ($\sim 10^{-6}$ Torr), mean free path for evaporated copper atoms is of the order of tens of metres, far exceeding d_{ss}. Hence the vapor travels ballistically along a straight line of sight from the boat to the substrate rather than diffusing.

d_{ss} also reflects a trade-off between film uniformity and deposition efficiency, since the flux reaching the substrate falls off approximately as the inverse square of the distance, $I \propto 1/d_{ss}^2$. A larger separation ($\sim$ 30–40 cm) improves thickness uniformity across the substrate but lowers the deposition rate and wastes more charge material. A smaller separation ($\sim$ 10–15 cm), as used here, increases the rate and conserves material. However, too small a separation cana lead to greater radiative heating of the substrate and increased non-uniformity near the edges of larger substrates.

2.3 Necessity of a Two-Stage Pumping System

A rotary vane pump alone cannot provide the vacuum required for a clean copper film. Rotary pumps reach only 10^{-2}–10^{-3} Torr, at which Eq. 1 gives a mean free path of only a few cm, well below d_{ss}; evaporated copper atoms therefore scatter off residual gas molecules, losing energy before reaching the substrate. Copper is also highly reactive toward residual oxygen and water vapor, and at 10^{-3} Torr the background flux of these species is large enough to oxidize a significant fraction of the depositing flux, yielding Cu_2O/CuO rather than metallic copper and degrading the film's conductivity. Gas molecules adsorbed on the substrate surface can further impede nucleation, promoting poor adhesion and film stress.

A turbomolecular pump is therefore used to lower the pressure, extending the mean free path to tens or hundreds of metres and restoring ballistic, line-of-sight transport of the vapor. The two pumps operate by different mechanisms over non-overlapping pressure ranges (Table 1)

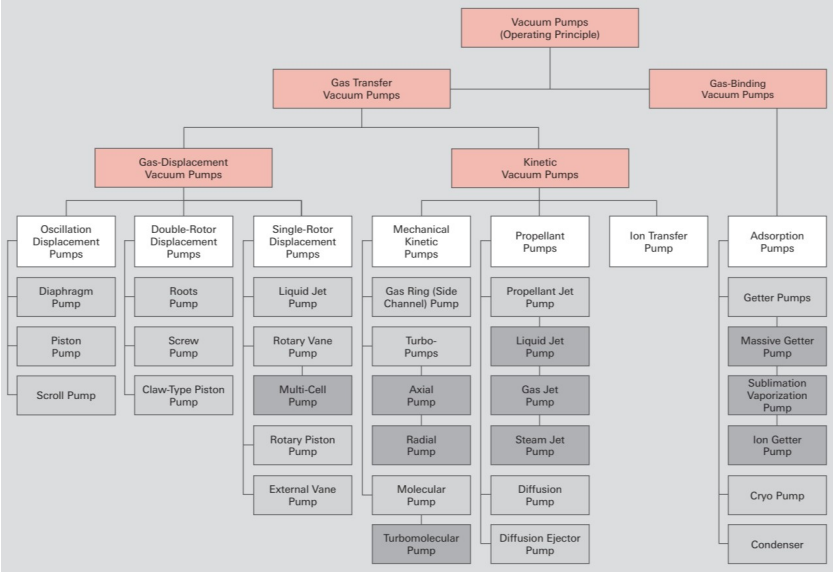


Figure 1: Overview of Vacuum Pumps [Image Courtesy: Pfeiffer- Vacuum Technology Book II]

Table 1: Comparison of the rotary vane (roughing) pump and the turbomolecular (high-vacuum) pump.

Feature	Rotary Vane Pump	Turbomolecular Pump
Working principle	Positive displacement: an offset rotor with spring-loaded vanes sweeps and compresses trapped gas volumes to the exhaust.	p transfer: spinning rotor blades collide with gas molecules, imparting momentum toward the exhaust via stationary stator blades.
Operating range	atm pressure (760 Torr) down to $\sim 10^{-3}$ Torr.	10^{-3} Torr down to 10^{-10} Torr.
Exhaust condition	Can exhaust directly to atmosphere.	Cannot exhaust to atmosphere; requires a foreline pressure below $\sim 10^{-1}$–10^{-2} Torr.
Gas flow regime	Viscous/continuum flow.	Molecular flow.
Rotor speed	1,400–1,800 rpm.	30,000–90,000 rpm.
Primary function	Roughing the chamber from atmosphere; backing the turbo pump's exhaust.	Lowering base pressure for ballistic vapor transport and film purity.
Safety constraint	Gas ballast avoids water-vapor condensation in the oil; low mechanical risk.	Cannot be exposed to atmosphere while it is running; overload shatters the rotor.

2.4 Pirani and Penning Gauges

Table 2: Comparison of the Pirani and Penning gauges.

Feature	Pirani Gauge	Penning Gauge
Category	Thermal-conductivity gauge.	Cold-cathode ionization gauge.
Working principle	A heated filament forms one arm of a Wheatstone bridge; as pressure drops, fewer gas collisions carry heat away, raising filament temperature and resistance.	A high voltage (1–5 kV) between anode and cathode sustains a glow discharge; a magnetic field lengthens electron paths, and the collected ion current is proportional to pressure.
Operating range	10^2 Torr down to $\sim 10^{-3}$ Torr.	10^{-3} Torr down to $\sim 10^{-8}$ Torr.
Role in this system	Monitors roughing/foreline pressure.	Monitors high-vacuum chamber pressure once below $\sim 10^{-3}$ Torr.
Hot filament	Yes, continuously heated.	No; uses a strong magnetic field and high voltage instead.
Gas dependence	Sensitive to the thermal conductivity of the gas species.	Sensitive to the ionization cross-section of the gas species.
Behaviour at atm.	Operates safely at 760 Torr.	Can't operate; Arcing or electrode damage.
Maintenance	Sensitive to oil-film condensation on filament.	Prone to cathode sputtering; requires periodic cleaning.

2.5 Resistive Heating of the Source

The charge material is held in a boat (tungsten, chosen for its high melting point and low vapor pressure at the evaporation temperatures of common charge metals) through which an AC current I is passed. The boat acts as a resistive heating element, and the power dissipated is

$$P_{\text{boat}} = I^2 R \tag{2}$$

where R is the resistance of the loaded boat. The boat temperature (hence vapor pressure and evaporation rate of charge) is controlled indirectly by ramping the current supplied through a variac-controlled step-down transformer. The current must be increased gradually to avoid thermal shock to the boat and to prevent spitting of charge material, which would produce an uneven film.

2.6 Film Thickness Monitoring

Film thickness and deposition rate are measured using a quartz crystal microbalance (QCM). A quartz crystal is driven at its resonant frequency; as material condenses on the crystal, the added mass lowers the resonant frequency according to the Sauerbrey relation,

$$\Delta f = -\frac{2 f_0^2}{A\sqrt{\rho_q \mu_q}}\,\Delta m \tag{3}$$

where f_0 is the fundamental resonant frequency of the crystal, A is the piezoelectrically active area, ρ_q and μ_q are the density and shear modulus of quartz, and Δm is the deposited mass. The STM2 controller

converts the measured frequency shift into a real-time deposition rate (Å/s) and cumulative thickness (Å). The density and acoustic impedance of the charge material have to be entered manually into the software.

2.7 Pumping Configuration

Two pumping configurations are referred to below:
(i) *Roughing*: rotary pump evacuates the chamber directly (chamber → roughing valve → rotary pump → atmosphere) while the turbo pump's gate valve stays closed to protect its rotor from atmospheric turbulence
(ii) *Backing*: rotary pump evacuates the turbo pump's exhaust (chamber → gate valve → turbo pump → backing valve → rotary pump → atmosphere), since the turbo pump compresses gas from $\sim 10^{-6}$ Torr at its inlet to only $\sim 10^{-2}$ Torr at its outlet and cannot exhaust directly to atmosphere (Section 2.3).

3 Experimental Setup and Procedure

3.1 Apparatus

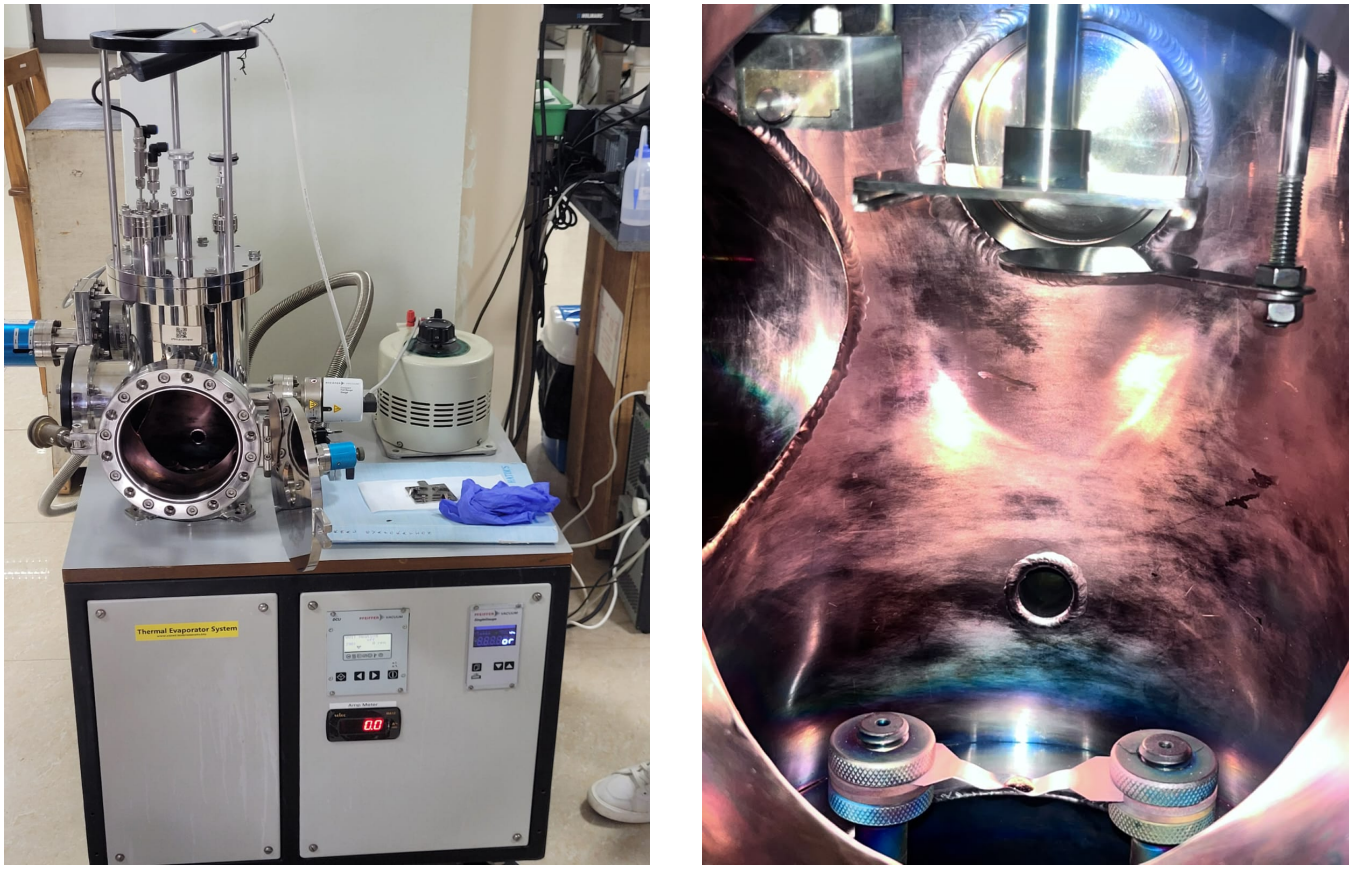


Figure 2: Thermal Evaporator System and the Vacuum Chamber

The deposition system consists of the following principal components:

- A stainless-steel vacuum chamber with a hinged door, gate valve, and vent valve.
- A two-stage pumping system: a rotary (backing) pump for preliminary vacuum ($\sim 10^{-3}$ mbar) and a turbomolecular pump for base vacuum ($\sim 10^{-5}$ mbar), connected via backing and roughing valves.

These pumps are compared in Table 1, and the rationale for using both is discussed in Section 2.3.

- A Pirani gauge, monitoring roughing/foreline pressure, and a Penning gauge, monitoring high-vacuum chamber pressure once below $\sim 10^{-3}$ Torr; the two gauge types are compared in Table 2.
- A tungsten boat source, resistively heated via a variac-controlled step-down transformer (Variac1, 0–230 V primary, 0–5 V secondary) with an in-line AC ammeter.
- A quartz crystal thickness monitor (QCM) interfaced with STM2 software, reporting real-time deposition rate and cumulative thickness.
- A glass microscope slide substrate, cleaned prior to loading.
- Charge material (copper).

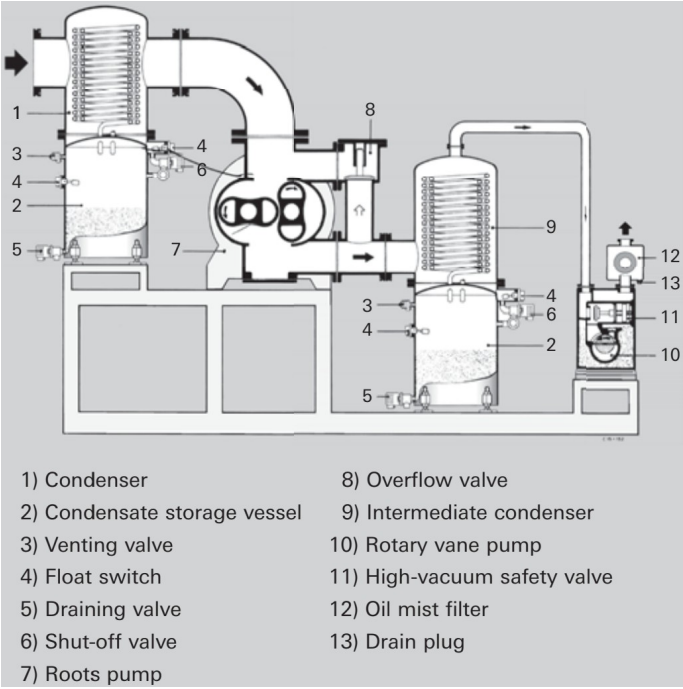


Figure 3: Schematic diagram: thermal evaporation deposition system [1]

3.2 Procedure

3.2.1 Chamber Evacuation

With all valves and the chamber closed, and with the glass substrate and charge material already loaded into the boat, the chamber was evacuated as follows:

1. The system power and the backing (rotary) pump were switched on.
2. The backing valve, connecting the rotary pump to the turbo pump, was opened.
3. The turbomolecular pump was started and allowed to reach full rotational speed.
4. Once the turbo pump reached full speed, the backing valve was closed and the roughing valve (between the rotary pump and the chamber) was opened; the chamber was evacuated to at least 2×10^{-1} mbar.
5. The roughing valve was then closed and the backing valve reopened, restoring the backing line to the turbo pump.

6. The gate valve, isolating the turbo pump from the chamber, was opened only once the pressure differential across it was below 1 mbar, and the chamber was pumped to base vacuum (at least 2×10^{-5} mbar) before proceeding to deposition.

If a pressure of 2×10^{-1} mbar is not reached within 5 mins of opening the roughing valve, a leak in the chamber or associated seals is possible, and the gate valve procedure would be used to continue pumping down while the leak is investigated. Throughout this sequence, the backing valve is never left closed for more than ~ 5 mins while the turbo pump was running, to avoid overloading the turbo pump's foreline.

3.2.2 Deposition

1. With base vacuum established, the STM2 control laptop and software were switched on, the sensor serial number entered, and data acquisition started.
2. The boat current was increased gradually using Variac1, and the corresponding AC current was monitored on the ammeter display.
3. Once a stable deposition rate was observed on the STM2 software, the shutter was opened to begin film growth on the substrate.
4. Rate (Å/s) and cumulative thickness (Å) were recorded at regular intervals until the target thickness was reached.
5. On reaching the target thickness, the shutter was closed, the variac was returned to minimum, and the transformer power was switched off.
6. The boat and chamber were allowed to cool for at least 5 minutes before venting.
7. The gate valve was closed slowly; the characteristic increase and sudden release of resistance felt at the valve indicated full closure.
8. The turbo pump was set to standby, and the chamber was vented and opened to remove the coated substrate, after which the vent valve was closed.

Upon completion of deposition and sample removal, the system was returned to a safe standby state by closing the door, cycling the roughing and backing valves to re-establish vacuum, and finally opening the gate valve to leave the chamber under vacuum ($\geq 10^{-4}$ mbar) with the turbo pump and rotary pump switched off. The turbo controller (DCU) was kept powered until the turbo pump speed had decayed to below approximately 10,000 rpm.

3.3 Sources of Uncertainty

The principal sources of systematic and random uncertainty in this experiment are:

- **Current/rate control**: manual ramping of Variac1 introduces run-to-run variation in the deposition rate, since current-to-rate relationship is not perfectly linear or reproducible between depositions.
- **Vacuum level fluctuations**: small leaks or outgassing during deposition can raise the chamber pressure, shortening the mean free path and altering the effective deposition rate and film uniformity.
- **Boat-to-substrate geometry**: variation in the source-to-substrate distance and angle across the substrate area produces a non-uniform film thickness (cosine-law falloff), which is not captured by the single-point QCM reading.
- **Gate valve timing**: pressure differential across gate valve at the moment of opening/closing was judged tactilely, introducing some variability in the exact vacuum level at each step.

4 Pressure Book-keeping

Table 3: Chamber pressure during evacuation

Time (min)	Stage	Pressure (mbar)
0	Roughing (rotary)	$\sim 1 \times 10^{0}$
t_1	Roughing target reached	2×10^{-1}
t_2	Turbo/gate valve, base vacuum	$\leq 2 \times 10^{-5}$

5 Results and Analysis

5.1 Deposited Cu Thin Film

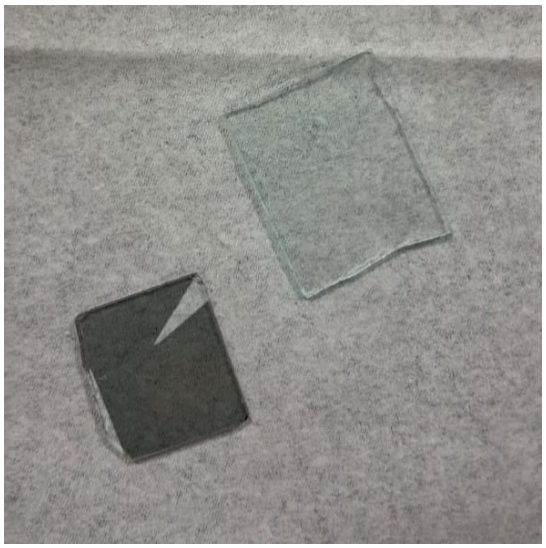


Figure 4: The Cu-Film deposited over glass slide, being compared against a clean glass slide

5.2 Fitting and Error Propagation

Where the deposition rate was approximately constant over an interval, thickness versus time was fit to a linear model,

$$t(\tau) = \bar{r}\,\tau + t_0, \tag{4}$$

with $\bar{r}$ obtained as the slope of a least-squares fit to the logged (time, thickness) pairs. The uncertainty in the final thickness, σ_{t_f}, combines the QCM's intrinsic reading uncertainty with the fit uncertainty in $\bar{r}$ propagated over the deposition duration τ_f:

$$\sigma_{t_f} = \sqrt{\sigma_{\mathrm{QCM}}^2 + (\tau_f\,\sigma_{\bar{r}})^2}. \tag{5}$$

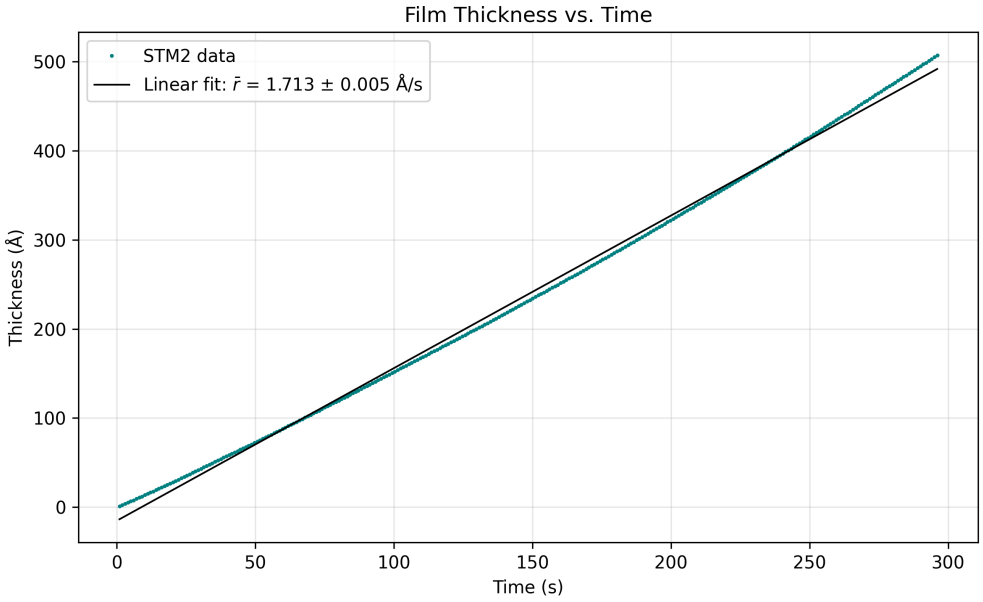


Figure 5: Film thickness versus time with a linear least-squares fit (Eq. 4) giving $\bar{r} = 1.713 \pm 0.005$ Å/s.

The mean deposition rate was measured as $\bar{r} = 1.7134, \text{Å/s}$ with an uncertainty of $\sigma_{\bar{r}} = 0.0046, \text{Å/s}$. The fit intercept was determined to be $t_0 = -15.429, \text{Å}$.

5.3 Result Summary

The STM2 software converts the raw QCM frequency shift into rate and thickness internally via the Sauerbrey relation (Eq. 3), so no separate unit conversion of the deposition data was required. The logged rate values were averaged over the steady-state portion of the deposition (excluding the initial ramp-up transient) to obtain a mean deposition rate $\bar{r}$, and the final cumulative thickness reading at shutter closure was taken as the measured film thickness t_f.

Table 4: Summary of measured deposition parameters.

Quantity	Measured Value
Mean deposition rate, $\bar{r}$ (Å/s)	1.7134
Final film thickness, t_f (Å)	-15.429

5.4 Thickness and Frequency Evolution

Figure 5 shows the STM2-logged film thickness as a function of time for the full deposition run, together with a least-squares linear fit of the form of Eq. 4. The fit returns a mean deposition rate

$$\bar{r} = 1.713 \pm 0.005 \text{ Å/s}, \tag{6}$$

with fit intercept $t_0 = -15.4$ Å. The data track the linear fit closely but show a small systematic upward curvature relative to it (measured thickness lies below fitted line for $t \lesssim 150$ s and above it for $t \gtrsim 250$ s).

Figure 6 shows the corresponding raw QCM crystal frequency over the same interval. The frequency decreases monotonically and smoothly by 402.6 Hz over the 296 s run, indicating that the crystal signal itself was stable and that the thickness curve of Figure 5 is a faithful Sauerbrey-relation (Eq. 3) conversion of the underlying frequency shift rather than an artifact of the STM2 processing.

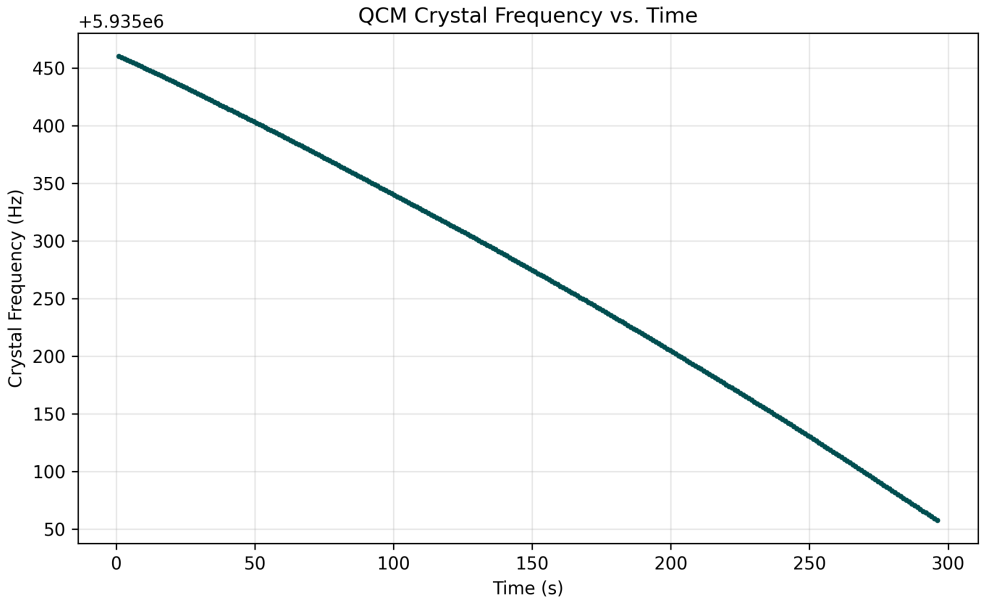


Figure 6: QCM crystal resonant frequency versus time showing a monotonic decrease over the deposition.

6 Discussion

The chamber reached the required base vacuum before the boat was heated (boat should never be heated without a vacuum better than 3×10^{-5} mbar); heating against poor vacuum risks oxidation of the charge and rapid burnout of the tungsten boat. The gradual current ramp on Variac1 is intended to bring the boat to evaporation temperature without thermal shock, which would otherwise crack the boat or cause the charge to spit, producing a non-uniform or particulate-laden film.

Agreement between the measured deposition rate/thickness and the target values should be assessed against the propagated uncertainty of Eq. 5; a deviation greater than a few σ would point to miscalibration of the QCM's material parameters (density, Z-factor) rather than to a genuine change in the physical deposition process. The dominant source of uncertainty in this system is expected to be the QCM calibration and the non-uniformity of the deposition geometry, both systematic in nature, rather than statistical scatter in repeated readings.

Any anomalies observed during the run, for example a stall in thickness reading or an unexpected pressure rise, are most plausibly attributed to a partial obstruction of line-of-sight between boat and substrate, a transient vacuum excursion, or spitting of the charge material during rapid heating. For a repeat experiment, tighter control of the current ramp rate, in-situ verification of the QCM's Z-factor for the specific charge material used, and rotation of the substrate holder to average out geometric non-uniformity would each be expected to reduce the overall uncertainty.

A Google Colab Notebook (Plotting and Curve Fitting)

The Google Colab Python notebook, which was used for plotting and fitting, can be found here.

References

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